# Human Activity Recognition with Missing-Joint Recovery

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**A measured extension of a published skeleton-action model.** We investigate whether limited repair of temporarily missing pose joints helps an unchanged pretrained classifier, and whether calibrated rejection makes accepted predictions more reliable.

## Base paper and verified journal status

**Continual Spatio-Temporal Graph Convolutional Networks**
Lukas Hedegaard, Negar Heidari and Alexandros Iosifidis.
*Pattern Recognition* 140 (2023), 109528. Published paper.

**Q1 evidence:** SCImago/SJR **2024**, category **Computer Vision and Pattern Recognition**; journal SJR 2.058, Q1. This verifies that named ranking/year, not 2023 JCR status. SCImago category listing.

## The implementation in one pipeline

Real OpenPose skeleton sequence > optional bounded causal repair > authors' fixed CoST-GCN* classifier > optional calibrated rejection.

The model combines information across body joints and across time. The paper's continual formulation reuses intermediate calculations during streaming. Our measurements use its weight-compatible clip path, with a recorded numerical check against the continual definition.

## What is ours

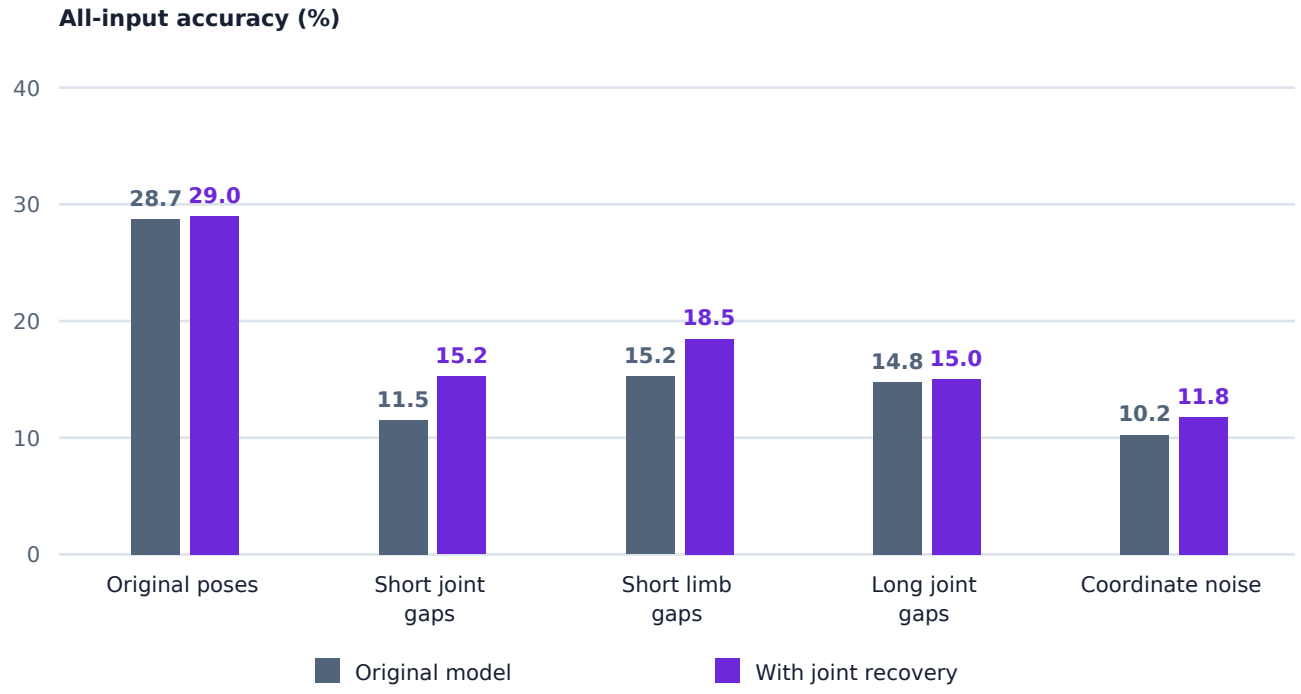
We implement pose repair, confidence decay, bounded gap handling, rejection calibration, controlled corruption experiments, ablations and a reviewable comparison. The pretrained classifier and its weights are credited to the authors; they were not trained by this project.

| Frozen setting | Measured configuration |
|---|---|
| Calibration-selected recipe | confident_hold_5 |
| Repair behavior | Confidence-gated hold (beta = 0) |
| Gap and update parameters | Maximum 5 frames; alpha 1; beta 0 |
| Prior confidence / decay | Minimum repair confidence 0.5; decay 0.9 |
| Coordinate guards | Speed <= 0.08; displacement <= 0.2 (input units) |
| Person-slot reset guard | Torso jump > 0.15 input units resets prior state |

Measured run: 2026-10-11T23:17:40.737167+05:30. 200 calibration clips; 400 disjoint test clips; 400 classifier outputs.

# Recovery gains and individual regressions

The primary contrast isolates recovery without rejection. Both methods see the same held-out clips, the same preprocessing and the same added corruption masks. All five conditions are retained.




| Condition | Difference | Paired 95% interval | Fixed / regressed |
|---|---|---|---|
| Original poses | +0.25 pp | -1.50 pp to +2.00 pp | 7 / 6 |
| Short joint gaps | +3.75 pp | +1.50 pp to +6.25 pp | 21 / 6 |
| Short limb gaps | +3.25 pp | +0.75 pp to +5.75 pp | 20 / 7 |
| Long joint gaps | +0.25 pp | -1.50 pp to +2.00 pp | 7 / 6 |
| Coordinate noise | +1.50 pp | +0.00 pp to +3.00 pp | 8 / 2 |

Differences are percentage points, not relative percentage improvements. A paired bootstrap resamples clip pairs; intervals crossing zero do not establish a consistent gain in this pilot. Corrected and regressed clips are both shown.

**Noise stress test:** recovery changes accuracy by +1.50 pp. The paired interval includes zero, so a consistent improvement is not established here. Gap repair is not a general-purpose denoising solution.

# Read accuracy alongside coverage

All-input accuracy and macro-F1 count abstentions as errors. Accepted accuracy uses only clips receiving a prediction. A higher accepted accuracy at lower coverage is a tradeoff, not proof of a better classifier.

| Condition | Variant | All-input accuracy | Macro-F1 | Accepted accuracy | Coverage |
|---|---|---|---|---|---|
| Original poses | Original model | 28.75% | 27.33% | 28.75% | 100.00% |
| | Joint recovery | 29.00% | 27.98% | 29.00% | 100.00% |
| | Rejection only | 28.50% | 27.98% | 33.93% | 84.00% |
| | Recovery + rejection | 28.75% | 28.42% | 33.82% | 85.00% |
| Short joint gaps | Original model | 11.50% | 8.78% | 11.50% | 100.00% |
| | Joint recovery | 15.25% | 12.66% | 15.25% | 100.00% |
| | Rejection only | 11.50% | 9.15% | 15.03% | 76.50% |
| | Recovery + rejection | 15.00% | 12.64% | 19.11% | 78.50% |
| Short limb gaps | Original model | 15.25% | 13.43% | 15.25% | 100.00% |
| | Joint recovery | 18.50% | 16.34% | 18.50% | 100.00% |
| | Rejection only | 14.50% | 13.31% | 18.18% | 79.75% |
| | Recovery + rejection | 17.75% | 16.44% | 21.98% | 80.75% |
| Long joint gaps | Original model | 14.75% | 13.21% | 14.75% | 100.00% |
| | Joint recovery | 15.00% | 12.95% | 15.00% | 100.00% |
| | Rejection only | 14.50% | 13.09% | 18.53% | 78.25% |
| | Recovery + rejection | 14.75% | 13.12% | 18.44% | 80.00% |
| Coordinate noise | Original model | 10.25% | 8.92% | 10.25% | 100.00% |
| | Joint recovery | 11.75% | 10.49% | 11.75% | 100.00% |
| | Rejection only | 10.00% | 8.88% | 11.53% | 86.75% |
| | Recovery + rejection | 10.75% | 9.47% | 12.57% | 85.50% |

Macro-F1 averages the 251 true classes represented in the test subset. The model retains all 400 outputs. 2 empty test clips are retained.

Frozen rejection thresholds: original inputs 0.080; repaired inputs 0.080. Threshold selection seeks the highest calibration accepted accuracy subject to at least 80% calibration coverage; this does not guarantee that coverage on unseen conditions.

# What this experiment establishes

- Original poses: accuracy difference +0.25 pp; interval includes zero.
- Short joint gaps: accuracy difference +3.75 pp; paired interval excludes zero.
- Short limb gaps: accuracy difference +3.25 pp; paired interval excludes zero.
- Long joint gaps: accuracy difference +0.25 pp; interval includes zero.
- Coordinate noise: accuracy difference +1.50 pp; interval includes zero.

## Measured processing cost

| Condition | Original ms/clip | Recovery ms/clip |
|---|---|---|
| Original poses | 31.91 | 47.15 |
| Short joint gaps | 30.87 | 43.58 |
| Short limb gaps | 31.04 | 43.68 |
| Long joint gaps | 31.06 | 43.49 |
| Coordinate noise | 30.94 | 43.79 |

Mean synchronized model processing time per 300-frame clip in batches; recovery runs on CPU. Excludes transfers, camera pose estimation, I/O and browser. Not per-frame streaming latency.

## Limits that must stay in the presentation

- Pose coordinates and labels are real Kinetics data. Joint masking and coordinate noise are synthetic perturbations; physical camera occlusion has not been validated.
- This is a sampled pilot, not a reproduction of the full paper benchmark or a state-of-the-art claim. Upstream validation data may have influenced the authors' checkpoint selection.
- Confidence-ranked person slots may change identities. Long gaps, abrupt motion and person swaps remain difficult. The optional identity guard is not a full tracker.
- The new 400-class classifier has no validated webcam integration. The existing MediaPipe six-activity dashboard is a separate demo, not this research checkpoint.

# A result you can inspect and explain

## Reproduction

Clone the project repository and install its recorded research dependencies. Setup downloads a pinned upstream checkout and verifies the checkpoint hash. The data utility downloads a fixed real-data subset using byte ranges rather than the full archive.

```
python -m pip install -r requirements-research.txt
python scripts/setup_research.py
python scripts/research_data.py
python -m research.benchmark --phase all
python scripts/render_research_report.py
```

The Colab notebook is notebooks/Research_Comparison.ipynb. Keep comparison.json, predictions.csv, calibration.json, design.json and the data manifest together. Selection is deterministic; never choose test clips based on whether predictions succeed.

## Short viva answers

**Why a graph network?** A body is naturally a graph of connected joints. Spatial layers combine neighboring joints; temporal layers learn movement across frames.

**What is the contribution?** A bounded repair and rejection pipeline plus a controlled, reproducible ablation study around the authors' unchanged pretrained classifier.

**Why is repair causal?** It uses previous and current measurements only. Filling a gap using a future observation would violate the streaming constraint.

**Why not compare our number directly with the paper?** The paper uses full benchmark splits. This smaller pilot and its added corruptions answer a different, explicitly scoped question.

**Why show negative results?** They identify the method's limits. A useful experiment must show regressions and unchanged conditions as well as improvements.

## Provenance and evidence

Numerical clip/continual check: passed. Maximum logit differences: clip vs continual 2.86e-06; clip vs step 4.77e-06. This is a check on the recorded calibration input, not validation of every possible stream.

Published base paper | Authors' implementation | SCImago 2024 category evidence

Upstream commit: a8fe2937a33f24cce65c1f8c2fc41081bceda721
Checkpoint SHA256: bec5c9563cc017a6e4739c2255ad76b24a046e445210b8fc965816435229452f
Comparison JSON SHA256: 346808497b94c5ee389f38629a4e46429b4895611430b73fae048e29643986b9